

Modeling Climate Anomalies with Statistical Analysis

Today's goals

Module 1

Introduction to Python for Data Analysis

Course Introduction

-  Course Updates and Accessibility Support
Reading • 1 min
-  Earn Academic Credit for your Work!
Reading • 10 min
-  Course Support
Reading • 10 min
-  Introduction to the Course
Video • 2 min
-  Meet the Instructor
Video • 1 min
-  Introduction to Pandas for Data Loading and
Exploration
Video • 25 min
-  Pandas
Reading • 15 min
-  Course Assignments with Jupyter
Reading • 10 min

Visualizing Data with Python

- Introduction to Matplotlib and Seaborn for Data Visualization
Video • 25 min

Module Assignment

- Introduction to Pandas and Matplotlib
Programming Assignment • 30 min
- Pandas and Matplotlib
Graded Assignment • 15 min

Module 2

Collecting Climate Data

Module 3

Visualizing & Analyzing Climate Data

Course Assignments with Jupyter

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For assignments in this course, you will need to set up your Python environment on your own device. This involves installing Jupyter Notebooks and several essential libraries: Matplotlib, Pandas, NumPy, Requests, and DataRetrieval. These tools will be fundamental as you progress through the course, enabling you to perform a wide range of data manipulation and visualization tasks.

Why Jupyter Notebooks?

Jupyter Notebooks are an open-source web application that allows you to create and share documents containing live code, equations, visualizations, and narrative text. They are widely used in data science for their ease of use and versatility, providing an interactive environment where you can write and execute code in real-time. This makes it easier to test hypotheses, visualize data, and document your workflow all in one place.

Essential Libraries

1. **Matplotlib:** A comprehensive library for creating static, animated, and interactive visualizations in Python. It's perfect for generating plots and graphs to visualize your data.
2. **Pandas:** A powerful data manipulation and analysis library. It provides data structures like DataFrames that are essential for handling structured data and performing operations such as merging, reshaping, and aggregating.
3. **NumPy:** A fundamental package for numerical computing in Python. It provides support for arrays and matrices, along with a collection of mathematical functions to operate on these data structures.
4. **Requests:** A simple and elegant HTTP library for Python, perfect for making API calls and handling HTTP requests. This will be crucial for retrieving data from various web services and APIs.
5. **DataRetrieval:** A library specifically designed for accessing hydrologic data from the US Geological Survey (USGS) and other sources. It simplifies the process of downloading and working with environmental data.

Getting Started

Follow these steps to set up your Python environment:

Download and Install Anaconda: Anaconda is a distribution of Python that includes Jupyter Notebooks and many other useful packages. You can download it from [Anaconda](#) (external link; opens in a new tab).

Create a New Environment: Open the Anaconda Prompt (or your terminal) and create a new environment:

```
1 conda create --name modeling_climate_anomalies python=3.10
2 conda activate modeling_climate_anomalies
```

Install Jupyter Notebooks:

```
1 conda install jupyter
2
```

Install the Required Libraries:

```
1 conda install matplotlib pandas numpy requests
2 pip install dataretrieval
```

Launch Jupyter Notebooks:

```
1 jupyter notebook
2
```

This will open the Jupyter Notebook interface in your web browser, allowing you to start creating and running notebooks.

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